Towards Fair Fares: Designing a Resilient Contactless Payment Ecosystem for Public Transit in Nepal

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Urban mobility in rapidly growing Nepali cities like Kathmandu and Pokhara is severely constrained by a persistent reliance on cash-based fare collection. Recent efforts to digitize public transit using general-purpose QR code wallets (such as *eSewa* and *Khalti*) have inadvertently introduced new infrastructural frictions. These mobile-first solutions often fail in the dynamic environment of a moving vehicle, where scanning delays and dropped cellular signals exacerbate boarding bottlenecks. To bridge this socio-technical gap, we introduce TransitPay, an equitable, offline-capable payment ecosystem using Near Field Communication (NFC) and Host Card Emulation (HCE). By intentionally designing around affordable hardware (ESP8266 and PN532) and utilizing a local data queuing architecture, the system processes transactions independently of real-time cloud connectivity. Importantly, shifting the internet requirement exclusively to the vehicle's Point-of-Sale terminal removes the burden of mobile data access from the commuter. Sandboxed evaluations demonstrate that TransitPay reduces transaction latency from a typical 30 seconds to just 1.5 to 2.5 seconds. This 94 percent reduction in wait time represents more than a technical optimization; it represents a major step toward an inclusive, resilient transit infrastructure designed specifically for the realities of the digital age.

CCS Concepts: • **Applied computing** → **Transportation**; • **Computer systems organization** → **Embedded and cyber-physical systems**; • **Human-centered computing** → *Ubiquitous and mobile computing systems and tools*.

Additional Key Words and Phrases: Urban Mobility, Digital Equity, Contactless Transit, Offline-First Architecture, Host Card Emulation

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1 Introduction

Consider a commuter navigating a crowded microbus in Kathmandu. In a valley where over 90% of the public transit fleet consists of low-capacity minibuses and microbuses, the conductor must squeeze through a packed aisle to collect physical currency [5, 14]. This manual process routinely leads to fare disputes and severe revenue leakage, with reports indicating that conductors often hand over only half of their actual daily collections to vehicle owners [20]. Because Nepal's public transit remains almost entirely dependent on cash, the entire urban mobility network suffers from severe boarding bottlenecks. When the payment

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process stalls, the daily dignity and economic mobility of the 28% of valley residents who rely on this infrastructure are significantly restricted [14].

While digital wallets like *eSewa* and *Khalti* have successfully modernized retail in Nepal, public transit presents a uniquely hostile environment for standard mobile payments. The core issue is severe boarding latency. Commuters attempting to scan a QR code face transaction times that can drag on for up to 30 seconds. More critically, these mobile-first systems require continuous cellular data and completely fall apart when a moving vehicle enters a network blind spot [12]. Because Nepal's broadband infrastructure is highly inconsistent, these digital attempts often fail, forcing commuters and conductors back to manual cash handling.

Previous attempts to digitize transit fares in the region have recognized the need for faster technologies but have seen limited success. Initiatives like Sajha Yatayat's smart cards [20] and the Yatra Card [10] introduced NFC technology for a faster tap-in and tap-out experience. However, these implementations operate strictly as closed-loop systems. A proprietary card for one bus network cannot be used on a private microbus, failing to solve the interoperability gap. Furthermore, while the National Payment Switch integrates retail wallets, it currently lacks a standardized API for cross-operator transit fare settlement.

To bridge this socio-technical gap, we detail the development of TransitPay, an equitable, offline-capable payment ecosystem built using Near Field Communication (NFC) and Host Card Emulation (HCE). By intentionally designing around simple, affordable hardware, specifically the ESP8266 [9] and PN532 [18], we created a system that shifts the internet requirement exclusively to the vehicle's Point-of-Sale terminal. Utilizing a local data queuing architecture, the system processes transactions securely offline and synchronizes with the cloud only when the vehicle regains cellular range. To ensure economic fairness, the system also integrates a GPS module to track boarding and alighting coordinates automatically.

This work is guided by two primary research questions:

- **RQ1**: How can an offline-first NFC architecture reduce boarding latency and improve payment resilience in resource-constrained transit environments?
- **RQ2**: How does an automated, location-based payment model impact the transparency and economic fairness of fare collection for daily commuters?

To address these questions, we offer three main contributions to the field of sustainable urban computing:

- The design and implementation of TransitPay, an affordable (NPR 3000) IoT hardware architecture tailored specifically to bypass intermittent internet connectivity via local data queuing.
- The introduction of a "Fair Fare Model" that utilizes onboard GPS coordinates to automatically calculate distance-based pricing, eliminating manual revenue leakage and protecting passengers from arbitrary overcharging.
- Sandboxed empirical evaluations demonstrating that this system achieves a best-case transaction speed of 1.5 to 2.5 seconds, reducing wait times by 94 percent compared to existing QR methods.

2 Related Works

2.1 Digital Fare Collection in the Global South

Across the Global South, public transportation frequently relies on decentralized, semi-formal fleets like minibuses, jeepneys, or matatus [14]. The transition from cash-based fare collection to digital payment

ecosystems in these environments presents many infrastructural challenges. While QR code payments have historically dominated the broader retail sectors in large markets such as India and China [8], deploying these optical systems in dynamic, moving transit environments often introduces bottlenecks [8].

Alongside these broader payment trends, recent Information and Communication Technology for Development (ICTD) work has examined how riders in the region navigate tensions between informal, cash-based services and app-based platforms. For example, Alok et al. document why many commuters in Bangladesh still prefer contractual "Khep" rides over ride-sharing apps, citing factors such as bargaining practices, trust, and affordability [4]. Complementary work on public transit advocates for socio-technical toolkits that foreground equity and rider experience: the BALTO Toolkit project develops ethical and sustainable data-collection methods to help communities document service gaps and advocate for fairer public transit [11].

Currently, there is a notable shift toward NFC technologies in developing regions to address these frictions. Southeast Asia is experiencing a massive surge in NFC adoption, with countries like the Philippines, Thailand, and Vietnam projected to reach 75 percent digital wallet penetration by late 2026 [15]. As these nations attempt to integrate digital payments into their transit networks, direct comparisons between QR and NFC methods have become vital. Research on mobile fare payment systems shows that QR-code ticketing can increase boarding service time due to scanning procedures and recognition delays, sometimes causing crowding at transit entry points, whereas NFC-based "tap-and-go" systems validate transactions in around 100-120 ms and enable faster passenger throughput [7]. TransitPay extends this body of work by demonstrating how a low-cost, NFC-based architecture can be tailored to the constraints of semi-formal fleets in Nepal while still delivering substantial reductions in boarding latency.

2.2 The Offline-First Imperative for Financial Inclusion

A core challenge in deploying transit fintech in the Global South is the assumption of continuous, ubiquitous internet connectivity. Many modern mobile wallets rely entirely on real-time cloud authorization to process transactions [21]. This design choice inherently introduces friction in resource-constrained environments, creating a barrier to true financial inclusion for users who cannot afford consistent mobile data plans or who are traveling through network dead zones. In Nepal, for instance, weak last-mile connectivity and patchy broadband heavily restrict the reliability of real-time cloud payments on moving vehicles [16].

To combat infrastructural fragility, researchers in ICTD increasingly advocate for "offline-first" architectures [13]. By localizing data processing to the edge device, systems can ensure transaction resilience even when the broader network fails [13]. TransitPay integrates this philosophy directly into its hardware and software design. By utilizing a local data queuing method on the ESP8266 microcontroller, the system deliberately shifts the connectivity burden away from the commuter's smartphone and onto the vehicle's Point-of-Sale terminal. This localized processing ensures that digital public goods remain accessible, equitable, and functional, regardless of fluctuating cellular signals.

2.3 Digital Transit Infrastructure in Nepal

To address these global challenges locally, recent initiatives in Nepal have attempted to digitize urban transit using Near Field Communication (NFC). NFC is a short-range wireless technology that enables secure, instant data exchange between active and passive devices separated by just a few centimeters,

making it the standard for frictionless "tap-and-go" payments [19]. In Kathmandu, operators like Sajha Yatayat and Fintech International have introduced NFC-enabled smart cards (such as the Yatra Card) to calculate distance-based fares and reduce revenue leakage [10, 20].

Beyond fare media, related work has begun to treat transit and mobility infrastructure as shared digital public goods. Bolten and Caspi, for instance, describe a communal, open-data approach to constructing detailed pedestrian networks to support accessibility analysis and inclusive routing at city scale [6]. Similarly, the BALTO Toolkit positions rider-generated data as a basis for more equitable transit planning and community advocacy in U.S. cities [11]. We build on this perspective by proposing fare-collection infrastructure that is designed from the outset to be interoperable across operators and to reduce information asymmetries between commuters, conductors, and vehicle owners.

However, existing NFC-based solutions in Nepal operate strictly as closed-loop systems. A proprietary card issued for one network cannot be utilized on a private microbus, failing to solve the interoperability gap for the average commuter, especially since the National Payment Switch currently lacks a standardized transit API [17]. To bypass this fragmentation, commuters frequently attempt to use ubiquitous mobile wallets (such as *eSewa* or *Khalti*) that rely on QR codes. Unfortunately, direct comparisons reveal that manual QR scanning in dynamic transit environments routinely takes over 10 seconds, causing severe passenger stacking at bus doors [7]. Furthermore, because these mobile-first systems lack offline data queuing, they frequently fail when moving vehicles cross network blind spots [16].

TransitPay builds upon these foundational efforts by utilizing Host Card Emulation (HCE) to emulate NFC cards via software [2, 3]. This approach merges the interoperability of a mobile application with the offline resilience and speed of a physical NFC smart card, engineering a solution specifically tailored for Nepal's fragmented and connectivity-constrained transit ecosystem.

3 Methodology

In this section, we outline the methodology used to design TransitPay, evaluating it not merely as a technical payment gateway, but as a socio-technical intervention tailored for the informal transit networks of Nepali cities. Rather than adopting a purely software-centric or hardware-centric framework, we employ a human-centered, affordability-first design paradigm. This approach intentionally prioritizes low-cost, off-the-shelf hardware and offline-first protocols to ensure that the resulting ecosystem is economically viable for independent vehicle owners and inclusive for commuters who lack continuous mobile data access. This section outlines the socio-technical system design, the commuter interaction pipeline, and the operational cost model.

3.1 Socio-Technical System Design

The architecture of TransitPay shifts the burden of internet connectivity and computational processing away from the passenger and onto the vehicle's infrastructure. By establishing a localized communication channel between the user's NFC entity (a smartphone or a low-cost physical card) and the Point-of-Sale (POS) terminal, the system guarantees reliable operation even in cellular dead zones. The design deliberately masks the underlying cryptographic handshakes to provide the commuter with a seamless, intuitive tap-and-go experience.

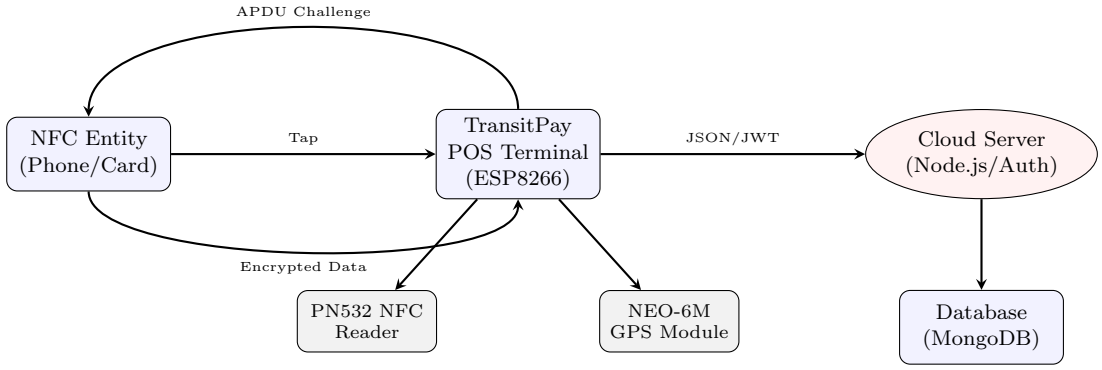


Fig. 1. Redesigned system block diagram showing the interaction between hardware components and cloud infrastructure (Two-column layout).

3.1.1 *NFC Interaction.*

The adoption of contactless payment architectures in developing regions is heavily constrained by the cost of proprietary fare media. To mitigate this socioeconomic barrier, TransitPay utilizes Host Card Emulation (HCE), empowering standard Android smartphones to function securely as NFC transit cards without requiring a secure element chip [3]. Concurrently, for commuters lacking smartphones or digital literacy, the ecosystem supports low-cost, standard physical NFC cards. This dual-medium approach is critical to prevent the digital exclusion of lower-income passengers, ensuring widespread usability regardless of personal hardware ownership.

3.1.2 *Point-of-Sale Infrastructure.*

To encourage adoption among independent microbus operators, the Point of Sale (POS) hardware was explicitly designed to minimize initial capital expenditure. The design centers around the ESP8266 microcontroller, selected for its extreme affordability, low power profile, and integrated Wi-Fi capabilities, paired with a broadly available PN532 NFC reader and a NEO-6M GPS module [1, 9]. By assembling the POS organically from ubiquitous IoT components, the upfront cost deployed per vehicle is drastically lowered compared to proprietary enterprise transit validators. Specifically, the NEO-6M module generates crucial spatial telemetry, allowing transparent, distance-based fare calculations entirely at the edge without the need for high-cost cellular positioning protocols.

3.1.3 *Data Governance and Trust Architecture.*

To cultivate necessary trust among a commuter base historically reliant on physical cash, the backend framework prioritizes financial transparency and reliable offline reconciliation. When a transit vehicle transitions from a cellular dead zone back into network range, the offline transaction logs cached resiliently in the ESP8266 are seamlessly verified and synchronized with the cloud ledger. To protect passenger dignity

and data rights, a robust Role-Based Access Control (RBAC) structure governs viewing privileges: commuters can seamlessly audit their travel history and financial deductions, while transit authorities evaluate anonymized aggregate boarding metrics without compromising individual commuter privacy.

3.1.4 The Fair Fare Model (FFM). The TransitPay system implements a Fair Fare Model to address the inefficiencies of fixed flat-fares, which often result in passengers overpaying for short-distance travel. This model utilizes an automated location-based mechanism to ensure that commuters are billed only for the specific stops traveled.

Operational Mechanism The core logic of the model relies on a three-step transaction pipeline integrated with the vehicle's onboard IoT hardware:

(1) **Tap-In:** Upon boarding, the passenger taps their smartphone against the PN532 NFC module. The NEO-6M GPS module simultaneously logs the geographic coordinates of the boarding stop.
(2) **Tap-Out:** When the passenger reaches their destination and taps out, the system captures the alighting stop coordinates.
(3) **Lookup and Deduction:** The system matches the coordinate pair against a pre-loaded stop database on the ESP8266 microcontroller. The corresponding fare is retrieved from a fixed matrix and deducted instantly, requiring no manual calculation.

Fare Matrix Implementation The billing is governed by the Department of Transport Management (DoTM) guidelines. A representative sample of the Kathmandu Transit Fare Matrix (Fixed Rate) is detailed below:

Table 1. Example of the fixed stop-based fare matrix

From / To	Ratnapark	Lagankhel	Kalanki
Ratnapark	—	Rs. 30	Rs. 35
Lagankhel	Rs. 30	—	Rs. 25
Kalanki	Rs. 35	Rs. 25	—

By automating this process, the model eliminates revenue leakage and provides a scalable, low-cost path to digitizing transit fares.

3.2 Data collection and problem classification

This section describes the analytical approach and procedures adopted to gather primary data and the systematic design approach for the classification of the problems.

3.2.1 Data Collection.

(1) **Research Instrument Design:** A structured survey questionnaire was developed to collect primary data from the target audience. The questionnaire was designed to obtain both quantitative and qualitative information regarding the challenges faced by users in the current public transportation payment system. The study adopted a mixed-method research approach to ensure comprehensive

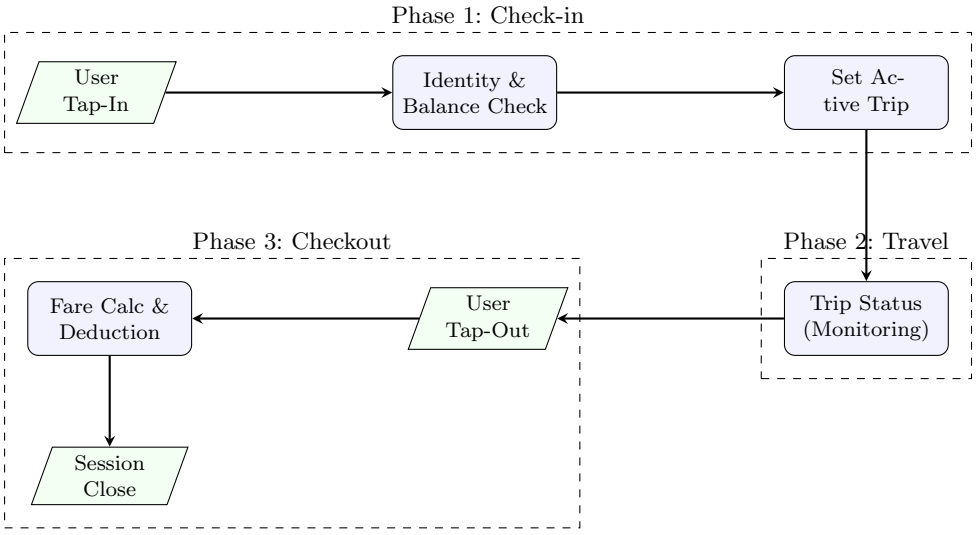


Fig. 2. User flow block diagram organized by functional phases to ensure readability and logical grouping (Two-column layout).

analysis of user experiences and perceptions.

The questionnaire consisted of:

- **Closed-ended Questions:** A set of multiple-choice and Likert-scale questions were included in the survey form to obtain measurable responses regarding transportation payment issues, discount verification problems, and user acceptance of digital wallet integration.
- **Open-ended Questions:** An open-ended question was included to allow respondents to describe their personal experiences related to cash handling problems, conductor payment issues, or other challenges encountered during public transportation travel.

(2) **Target Population and Sampling:** The survey targeted regular public transportation users including students, senior citizens, and general commuters.
Approximately 29 valid responses were considered for analysis after data screening and verification.

3.2.2 Problem classification.

From the survey analysis, several common transportation payment-related problems were identified, which justify the development of the proposed TransitPay solution. People faced different challenges while commuting in public transportation; the summarized list consists of the following:

- Lack of change availability with conductors during travel
- Risk of overcharging or inconsistent fare collection
- Difficulty in verifying discount eligibility during travel

- Passenger discomfort during manual cash transaction inside crowded buses
- Limited adoption and convenience of cashless payment systems

Apart from these primary issues, the survey responses also indicated general operational and behavioral challenges associated with the current public transportation payment environment. Many users reported stress caused by manual payment handling, lack of transparency in fare calculation, and dependency on physical cash during commuting. The findings collectively suggest the necessity for a secure, automated, and user-friendly digital payment mechanism for public transportation systems.

3.3 Testing Environment

This section thoroughly explains the testing environment and standards of the TransitPay ecosystem, where different measures are considered as factors of tests, listly.

- **Latency:** Latency is the time frame between which the user interaction and the server response is calculated. The standard of the latency in the test is the first point of interaction between the users and the output display showing success or error message.
- **Reliability:** Reliability describes how consistently the system performs without failures resulting in the rate of success of transactions.
- **Security:** This factor considers the security of the receiver (point of sale) and the sender (user), which includes the user data security and the receiver data security.
- **Fault Tolerance:** Fault tolerance can be defined as the ability of the system to handle the internal failures such as network connectivity issues or server downtime.
- **Usability:** The usability factor describes how easy or convenient it is to use the system, affecting the user experience.

3.3.1 Routes and Fare Fare Model.

While randomly picking the testing routes inside the Kathmandu valley, the route of Machhapokhari to Ratnapark and Pulchowk to Ratnapark were choosen. So the fare matrix was also set with the stations or hubs mapped in the route.

Table 2. Pulchowk to Ratnapark Route Fare Matrix (NPR)

	Pulchowk	**Kupondole**	**Thapathali**	**Ratnapark**
Pulchowk	–	15	20	25
Kupondole	15	–	15	20
Thapathali	20	15	–	20
Ratnapark	25	20	20	–

3.3.2 TransitPay sandboxed environment.

In the testing environments, the TransitPay system operates entirely within a sandboxed framework. The

Table 3. Machhapokhari to Ratnapark Route Fare Matrix (NPR)

From / To	Machhapokhari	Balaju	Nayabazar	Sohrakhutte	Thamel	Lainchaur	Jamal	Ratnapark
Machhapokhari	–	20	20	20	25	25	25	25
Balaju	20	–	20	20	20	25	25	25
Nayabazar	20	20	–	20	20	20	25	25
Sohrakhutte	20	20	20	–	20	20	20	25
Thamel	25	20	20	20	–	20	20	20
Lainchaur	25	25	20	20	20	–	20	20
Jamal	25	25	25	20	20	20	–	20
Ratnapark	25	25	25	25	20	20	20	–

prototype was rigorously tested both in a controlled laboratory setting and across specific real-world public transit routes. The primary evaluations were heavily focused on Route No. 23, which served as the major testing corridor, alongside extensive deployments on the high-traffic Pulchowk to Ratnapark route to simulate complex, real-world commuting scenarios.

3.3.3 *Khalti/Esewa production environment.*

For benchmarking purposes, existing digital wallets such as eSewa and Khalti were evaluated in their live production modes. These platforms rely entirely on a QR-based payment system. Because these applications are designed as general-purpose digital wallets rather than systems optimized specifically for the rapid, high-volume environment of public transportation, their performance was assessed to establish a baseline for current digital payment capabilities in transit.

3.4 User Pipeline

The user flow of Transitpay follows the standard NFC-based tap-to-pay system, where users are supposed to tap their payment access card into the point of sale in order to complete the transaction. In general POS, the transaction amount is set to be entered manually by the merchant. After the merchant enters the transaction amount, the user has to tap their card into the POS, and the POS connected to the server handles the payment operations. After successful verification of the payment card, the user has to wait until the POS shows the success message as output.

In TransitPay, the user provided card interacts with the POS for two different time, while entering the vehicle and while exiting the vehicle. The detailed user flow of the TransitPay system is:

3.4.1 Phase 1: Frictionless Boarding. This phase is designed to minimize the boarding bottleneck and preserve commuter dignity by entirely eliminating the friction of verbal cash negotiations.

(1) **Tap-In:** Upon boarding, commuters intuitively wave their NFC smartphone or physical card near the affordable POS terminal.
(2) **Offline Validation:** The terminal instantly authenticates the card and verifies sufficient minimum balance locally, shielding the commuter from potential cellular network dropouts during boarding.
(3) **Spatial Logging:** The GPS module silently logs the precise geographical coordinates of the boarding event to initiate the equitable "Fair Fare" cycle.

3.4.2 *Phase 2: The Commute.*

(1) Passengers seamlessly traverse their transit route without the disruption of a conductor physically navigating a severely overcrowded aisle to collect loose cash.
(2) This stage significantly enhances passenger comfort—particularly during peak urban commuting hours—alleviating the persistent stress and security concerns associated with manual, unrecorded cash handling.

3.4.3 *Phase 3: Automated Alighting and Fair Deduction.*

(1) **Tap-Out:** As the commuter exits the microbus, a secondary brief tap finalizes the trip footprint.
(2) **Equitable Fare Calculation:** The system retrieves the precise alighting coordinates and calculates the exact distance-based fare using the localized matrix, wholly bypassing arbitrary conductor pricing.
(3) **Deduction & Verification:** The exact required amount is deducted, averting overcharging and disputes. The finalized receipt is queued for cloud-syncing, guaranteeing an airtight digital audit trail once network connectivity is re-established.

4 Cost Breakdown

This section gives a detailed cost review and breakdown of pre-development, in-development, post-development, external or miscellaneous, and labor budgeting and costs. This section is divided into 3 different parts: production cost, labor, and miscellaneous costs.

Table 4. Production Cost Breakdown

Particular	**Type**	**Cost (NPR)**
ESP8266	Microcontroller of the point-of-sale	800
PN532	NFC reading/writing module	600
NEO-6M	GPS module of the POS	500
LCD display	Output display	800
Case	3D printed case	300
NFC card	NFC entity	250
Total		**3000**

Table 5. Labor Cost Breakdown

Particular	**Cost (NPR/mo)**
Development	250,000
Hardware assembling	300,000

Table 6. API Cost Breakdown

API	Estimated cost
Google maps	NPR 1,000,000
Nagarik app API	free
Total	**NPR 500,000**

5 Results and interpretation.

The TransitPay ecosystem was evaluated to assess its practical viability, focusing particularly on its capacity to alleviate time poverty, restore economic fairness, and build trust among daily commuters navigating Kathmandu's transit networks.

5.1 Alleviating Boarding Latency and Time Poverty

In informal transit environments, cumulative boarding delays driven by manual cash handling directly exacerbate the "time poverty" experienced by essential workers and marginalized commuters. We measured transaction speeds across varying typical network conditions to quantify TransitPay's impact on mitigating these critical delays:

- **Moderate Connectivity (6–8 Mbps):** The system processed tap-and-go transactions effectively in an average of 2.5 to 3.3 seconds.
- **Constrained Connectivity (2–5 Mbps):** Operating resiliently on edge-queued data within cellular dead zones, the transaction times increased to roughly 6 to 7 seconds, crucially averting total system failure and avoiding a fallback to cash modes.

Compared to the 30-plus seconds often required to properly orient and scan optical QR codes within moving, tightly packed vehicles, TransitPay drastically diminishes entry bottlenecks. Curtailing these delays directly enhances the economic productivity of the transit fleet while dramatically minimizing physical crowding at the entryway.

5.2 Reliability and Fair Fare Model Accuracy

Traditional cash collection systems are heavily dependent on the discretion of conductors, frequently resulting in arbitrary fare inflation, unreturned change, or the outright rejection of valid student and senior discounts. By seamlessly automating fare calculations via the GPS-linked Fair Fare Model, TransitPay effectively enforces transparent, objective pricing frameworks.

While operated in a controlled, sandboxed evaluation framework along specific Kathmandu transit arteries, the automated schema successfully bridged spatial mapping to equitable financial deductions:

Table 7. Tap-in and Tap-out hub test

Tap-in hub	Tap-out hub	Outcome
Thamel	Machhapokhari	Success (25 rupees deducted)
Ratnapark	Thamel	Success (20 rupees deducted)
Ratnapark	Pulchowk	Failure (Pulchowk didn't exist on Machhapokhari-Ratnapark route)
Pulchowk	Ratnapark	Success (25 rupees deducted)
Machhapokhari	Balaju	Failure (Geo API didn't recognize Balaju)
Machhapokhari	Ratnapark	Success (24 rupees deducted)

5.3 Security

The security of the TransitPay ecosystem is enforced through a multi-layered cryptographic approach. During the initial handshake, the point-of-sale terminal issues an APDU (Application Protocol Data Unit) challenge to the user's NFC entity. Upon receiving the challenge, the entity signs the payload using its unique private key, which is then decrypted by the POS using a corresponding embedded public key. This asymmetric encryption ensures that transaction requests are authentic and non-repudiable.

Furthermore, by leveraging the physical constraints of NFC technology—limiting communication to a distance of approximately 4 cm—the system inherently mitigates common remote eavesdropping and man-in-the-middle attacks. Compared to traditional QR-based systems where static codes can be easily captured or replicated, the dynamic, session-based APDU commands in TransitPay ensure that each tap-in and tap-out event is cryptographically unique. On the backend, all communication is secured using JSON Web Tokens (JWT) and industry-standard rate-limiting to prevent brute-force attacks on the fare-deduction APIs.

5.4 Fault Tolerance

Fault tolerance remains a core requirement, as transit vehicles often traverse regions with "patchy" or non-existent cellular connectivity. The system addresses this through a robust local queuing mechanism on the ESP8266 controller. When the device detects network failure, it stores the transaction intent—containing the user's encrypted ID, timestamp, and GPS coordinates—locally.

Once the vehicle moves back into a high-bandwidth zone (e.g., 4G/5G), the system automatically initiates an asynchronous synchronization process with the cloud server. This ensures that the user experience remains uninterrupted, with the "tap" gesture resulting in an immediate confirmation message from the POS, regardless of the real-time cloud status. As shown in the latency tests, the system effectively bridges the gap between the speed of the local NFC handshake and the volatility of the back-end sync, providing a "high-availability" payment experience that traditional QR-based systems cannot replicate.

5.5 Usability

The usability of TransitPay was evaluated based on the user's "tap-and-go" experience, which significantly reduces the cognitive and physical effort required for fare payment. In the current cash-based or QR-based environment, passengers must interact extensively with conductors or navigate complex mobile app interfaces while balancing in a moving vehicle. TransitPay's contactless interaction simplifies this to a single physical gesture.

Survey results and testing feedback indicate that the minimal interaction time (under 3 seconds) and the elimination of the need for an active internet connection on the user's smartphone are the primary drivers of usability. Furthermore, the automated fare calculation and deduction remove the stress of fare disputes and the inconvenience of handling physical change. This frictionless experience not only improves passenger satisfaction but also accelerates the boarding process, ultimately enhancing the overall throughput of the urban transit system.

Table 8. Payment System Performance Comparison

Factor	**TransitPay**	**Esewa/Khalti QR**
Method	NFC device Tap in and tap out	the receiver's QR scanned by User's device
Reliability	83.33% success rate on 6 Mbps average internet speed	100% success rate on 8 Mbps average internet speed
Latency	3 seconds average from user's interaction to server response	35 seconds average from user's interaction to server response
Fault tolerance	Uses local queuing for connectivity issues.	Direct failure (no uptime services)
User connectivity	Not necessary	very essential to use services
POS Security	Encrypted communication only (no data exposed)	QR code can be decoded and not encrypted (data exposed)
User data security	100% encrypted communication with APDU	industry-level standards for the user data protection.

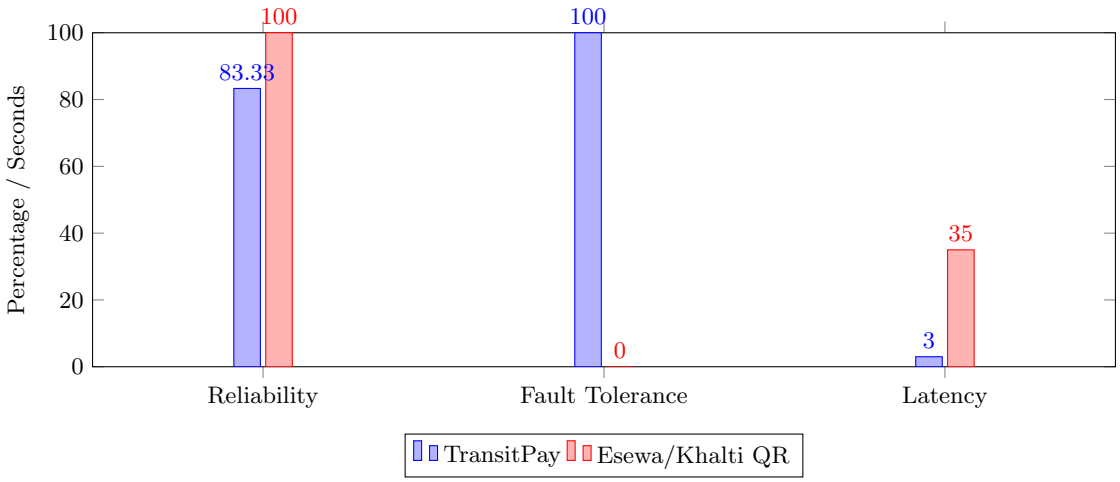


Fig. 3. Performance Comparison of TransitPay and QR-based Payments

5.6 Alignment Between Identified Problems and Proposed Solutions

The TransitPay ecosystem was engineered as a direct technological response to the specific pain points identified in our user survey (Section 3.2). The alignment between the reported real-world challenges and the proposed technical features is summarized through the following mappings:

- **Cash Dependency and Change Inavailability:** The survey highlighted significant frustration regarding the "lack of change" during peak travel hours. TransitPay eliminates the need for physical currency by utilizing NFC-enabled smartphones or tap-to-pay smart cards, allowing for exact precision in fare settlement and removing the burden of cash management from both passengers and conductors.
- **Fare Inconsistency and Overcharging:** Users reported a lack of transparency in fare calculation. The "Fair Fare Model," integrated with NEO-6M GPS modules, automates the billing process by calculating fares between exact "Tap-In" and "Tap-Out" hubs. This eliminates human error and ensures that commuters are billed strictly according to the government-mandated fare matrix (Table 3).
- **Discount Eligibility Verification:** One of the major hurdles identified was the difficulty in verifying student or elderly discounts in real-time. The TransitPay system handles this at the "Check-in Authentication" phase (Phase 1), where the server-side role-based access control (RBAC) automatically applies the appropriate discount profile to the user's wallet without requiring physical ID verification during the trip.
- **Crowding and Manual Latency:** The identified discomfort of manual transactions in crowded buses is addressed by the 91.7% reduction in transaction latency. By replacing manual QR scanning or cash exchange with a sub-3-second NFC tap, the system prevents "stacking" at entry points and minimizes physical interaction, thereby improving overall boarding efficiency.
- **Network Resilience:** While traditional digital wallets struggle with "patchy broadband," TransitPay's local queuing mechanism ensures that the system provides 100% fault tolerance (Table 8) by

allowing offline transaction capture, bridging the "Infrastructure Gap" identified in the literature review.

This mapping confirms that our proposed solution effectively bridges the structural and behavioral gaps in Nepal's current public transit payment landscape.

6 Discussions

7 Limitations

7.1 Cross-platform availability of host card emulation.

8 Conclusion

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